AR / VR DEVELOPMENT

From Fundamentals to Advanced Concepts

Unity • C# • Vuforia • AR Foundation • ARCore / ARKit

Purpose: A structured study guide for AR, VR, Unity development, Vuforia image-target tracking, markerless AR, and viva preparation.

LEARNING PATH

Step	Topic	Outcome
01	XR fundamentals	Understand AR, VR, MR, tracking and spatial computing.
02	Unity essentials	Scenes, GameObjects, Components, Transforms, prefabs and C#.
03	Marker-based AR	Vuforia, image targets, target quality and events.
04	Markerless AR	AR Foundation, ARCore / ARKit, planes, raycasts and anchors.
05	Advanced practice	Interaction, optimisation, builds, testing and troubleshooting.

Source basis: Your supplied Unit 3 lab PDF on building AR apps in Unity. General VR concepts and broader Unity explanations are supplemental study material.

QUICK DEFINITIONS

Term	Meaning
XR	Extended Reality: umbrella term for AR, VR and MR.
AR	Adds digital content to a view of the real environment.
VR	Immerses the user in a computer-generated environment.
MR	Blends real and virtual content with spatial awareness and interaction.
Tracking	Estimates the position and orientation of a device, target or object over time.

1. AR, VR, MR and XR Fundamentals

These terms are related but not interchangeable. Start by identifying how much of the real environment the user sees and how virtual content behaves.

Technology	Experience	Examples
AR	Real environment remains visible; digital content is overlaid.	AR filters, furniture preview, image-target textbook.
VR	Virtual environment replaces most or all of the real-world view.	VR training, virtual tours, immersive games.
MR	Real and virtual content are spatially aware and may interact.	Spatial collaboration, virtual objects aligned with real equipment.
XR	Umbrella term for AR, VR and MR.	Cross-reality applications.

Core AR capabilities

- **Motion tracking:** estimates device movement and rotation using camera imagery and inertial sensors.
- **Environmental understanding:** detects planes, feature points, depth or scene geometry where supported.
- **Light estimation:** estimates environmental lighting so virtual objects fit the scene.
- **Anchors:** associate virtual content with a tracked real-world pose.
- **Interaction:** lets users tap, select, move, rotate or inspect virtual content.

Marker-based vs markerless AR

Aspect	Marker-based	Markerless
Tracking cue	Known image / target.	Detected surfaces, visual features and environment cues.
Typical Unity tool	Vuforia Engine.	AR Foundation with ARCore / ARKit providers.
Example	3D model appears on a printed card.	Tap the floor to place a virtual chair.
Main consideration	Target must be visible and trackable.	Needs supported device capabilities and suitable surroundings.

2. Unity Fundamentals

Unity combines scenes, objects, components, scripts and assets into an application.

Concept	Meaning
Scene	Saved environment containing objects, lights, cameras and scene data.
GameObject	An object in a scene; components add its data and behaviour.
Component	A unit such as Transform, Camera, Collider, Renderer or a script.
Transform	Position, rotation and scale relative to the object's parent.
Material	Controls how a surface appears, including colour and shader properties.
Prefab	Reusable template for creating configured GameObject instances.
Hierarchy	Shows objects and parent-child relationships in the open scene.
Inspector	Displays and edits selected object's components.
Project window	Shows scripts, materials, prefabs and other assets.

C# lifecycle and key terms

Method / term	Purpose
Awake()	Initialise references or internal state when loaded.
Start()	Run setup before the first frame update.
Update()	Run per-frame logic; use Time.deltaTime for time-based movement.
FixedUpdate()	Physics-related updates on the fixed timestep when appropriate.
OnEnable / OnDisable	Subscribe or unsubscribe from events.
Instantiate / Destroy	Create or remove runtime objects.
Debug.Log	Write diagnostic messages to the Console.

Example: rotate a cube smoothly

```
using UnityEngine;

public class Rotator : MonoBehaviour
{
  public float speed = 90f;
  void Update()
  {
    transform.Rotate(0f, speed * Time.deltaTime, 0f);
  }
}
```

The script file should be named **Rotator.cs** to match the class. Save, wait for Unity to compile, and attach the script to the intended GameObject.

3. Vuforia Engine: Marker-Based AR

Vuforia supports target-based AR workflows. An image target is recognised from camera input, and virtual content is rendered relative to the target's tracked pose.

Image-target workflow

Step	Action
1. Prepare image	Use a sharp, detailed target with distinctive features.
2. Configure Vuforia	Set up the required licence and target database for the project.
3. Add AR Camera	Configure the Vuforia camera and licence key.
4. Add Image Target	Assign the correct target image and its physical width.
5. Add 3D content	Make the model a child of the Image Target; set sensible scale and position.
6. Test	Check tracking distance, angles, lighting and stability.

Important Vuforia concepts

- **Vuforia Engine:** SDK providing target tracking and related AR features.
- **AR Camera:** camera configured for Vuforia tracking; avoid conflicting extra cameras.
- **Image Target:** known image used to estimate the target's pose and scale.
- **Target events:** found/lost events can trigger audio, text or visibility changes.
- **Target hierarchy:** child content follows the target's tracked pose.
- **Physical width:** set the target's real printed width in the unit expected by the component, commonly metres.

How to design a trackable image

- Prefer sharp, high-contrast images with distinctive corners and detail spread across the image.
- Avoid blur, glare, smooth gradients, large blank regions and repetitive patterns.
- Use a flat matte print where possible; improve lighting and reduce reflections.
- Use Vuforia's target evaluation tools instead of guessing image quality.

Feature-point experiment from the Unit 3 lab

Image	Example	ORB preview count	Interpretation
A	Smooth gradient + logo	280	Few details outside the logo.
B	Checkerboard	544	Repetition may confuse matching.
C	Detailed collage	1,391	Distinctive, distributed detail.
D	Soft low-contrast image	0	No useful corners detected in this preview.

These values are from the lab's OpenCV ORB preview, not Vuforia's own detector. Exact counts can differ. High count alone does not guarantee good tracking; features should be distinctive and well distributed.

Target-found interaction example

When a target is found, show an information panel or play a short explanation. When the target is lost, hide the panel or pause the audio. Test repeated found/lost transitions and make sure references and event subscriptions are handled correctly.

4. AR Foundation: Markerless AR

AR Foundation provides common Unity APIs for AR features. Provider plug-ins supply platform-specific implementation: ARCore for supported Android devices and ARKit for supported Apple devices.

Architecture

Layer	Role
Unity AR application	C# scripts, 3D models, prefabs and UI.
AR Foundation	Common Unity AR APIs.
ARCore XR Plug-in	Provider for supported Android devices.
ARKit XR Plug-in	Provider for supported iPhone / iPad devices.

Core components

Component	Purpose
AR Session	Manages AR lifecycle such as start, pause and reset.
XR Origin (Mobile AR)	Defines the coordinate origin and contains the AR camera/tracking structure.
AR Plane Manager	Detects and manages trackable planes when supported and configured.
AR Raycast Manager	Tests a screen point against AR trackables such as planes.
AR Anchor Manager	Creates/manages anchors where supported by the provider.
Prefab	Reusable object template that can be instantiated at a detected pose.

Tap-to-place task flow

Camera tracking → Plane detection → User tap → Screen-space raycast → Plane hit pose → Instantiate prefab

Simplified logic sketch

```
if (userTappedScreen)
{
  if (raycastManager.Raycast(screenPosition, planeHits))
  {
    Pose pose = planeHits[0].pose;
    Instantiate(prefab, pose.position, pose.rotation);
  }
}
```

This is a conceptual sketch, not a complete drop-in script. A working script also needs valid component references, input setup, prefab assignment and an AR scene configured for the selected Unity/package versions.

ARCore vs ARKit vs AR Foundation

Technology	Role
ARCore	Google's AR platform for supported Android devices.
ARKit	Apple's AR framework for supported iPhone / iPad devices.
AR Foundation	Unity's common API layer over supported AR provider plug-ins.

5. VR Fundamentals

The supplied Unit 3 lab focuses mainly on AR. This section adds general VR concepts for broader AR/VR study.

- **Immersion:** the user views a virtual environment through a headset or another display setup.
- **Head tracking:** the view updates as the user's head moves.
- **Controllers / hand tracking:** support selection, grabbing, pointing and gestures.
- **Spatial audio:** direction and distance help locate virtual events.
- **Locomotion:** teleportation, smooth movement or room-scale walking.
- **Comfort:** stable frame rate, predictable movement and appropriate locomotion reduce discomfort.

Typical Unity VR workflow

Step	Action
1	Define user, goal, target headset and interaction model.
2	Build scene geometry, lighting, scale, materials and colliders.
3	Install required XR packages and enable the headset provider.
4	Set up the XR rig / Origin and input system for the chosen configuration.
5	Add controller or hand interactions and UI.
6	Test on-device for scale, comfort, tracking, performance and safety.

Concern	AR design	VR design
Environment	Real world remains visible.	Virtual world dominates view.
Tracking	Device pose plus image / plane / environment tracking.	Headset pose and controller / hand tracking.
Safety	Do not obscure real-world hazards.	Provide boundary awareness and safe play space.
Performance	Stable tracking and believable alignment.	Stable frame timing and low latency are critical.

6. Advanced Concepts and Optimisation

Concept	Meaning / why it matters
SLAM	Simultaneous Localisation and Mapping: estimating device pose while mapping the environment.
Feature points	Distinctive image locations used to match visual information across frames.
Plane detection	Finds surfaces such as floors and tables for placement.
Raycast	Tests a screen position or ray against trackables or scene geometry.
Anchoring	Associates content with a tracked pose so it remains aligned with a location.
Occlusion	Lets nearer real objects hide virtual content when supported by depth / scene understanding.
Light estimation	Helps match virtual object lighting to the real environment.
Spatial mesh	Represents real-world geometry for interaction or occlusion where supported.
XR Interaction Toolkit	Unity toolkit for common XR interaction patterns; setup depends on package version.

Performance and quality checklist

- Use appropriately sized textures and meshes; avoid unnecessary complexity.
- Avoid expensive work every frame unless required; cache component references.
- Use Time.deltaTime for frame-rate-independent movement.
- Test tracking in different lighting, angles and distances.
- Editor simulation is useful but does not replace real-device testing.
- Use logs and profiler data to investigate performance issues.
- For VR, prioritise stable frame timing and comfort; for AR, prioritise stable tracking and alignment.

Interaction design rules

- Make interactive objects visually distinct and easy to select.
- Give immediate feedback after a tap or selection.
- Provide a clear reset or remove-object action for placement demos.
- Avoid placing virtual objects over important real-world hazards.
- Use concise labels or optional audio instead of overwhelming the scene.

7. Setup and Troubleshooting

Vuforia checklist

- Install or import the appropriate Vuforia package for the project.
- Configure the licence key and target database as required.
- Set up the Vuforia AR Camera and add the Image Target.
- Assign the image; set its physical width; attach a correctly scaled child model.
- Test with a webcam or compatible device according to the chosen workflow.
- If tracking fails, check image quality, lighting, glare, target visibility and hierarchy.

AR Foundation checklist

- Install AR Foundation and the provider plug-in for the target platform.
- Enable the provider in XR Plug-in Management.
- Create AR Session and the appropriate XR Origin.
- Add AR Plane Manager and AR Raycast Manager for plane placement.
- Assign required plane/object prefabs and check input handling.
- Use XR Simulation if available, then build and test on a supported device.

Common problems

Symptom	Likely cause	Check
Unexpected camera view	Camera conflict or incorrect AR scene.	Camera hierarchy and provider setup.
Vuforia licence error	Wrong or incomplete key.	Correct key and configuration.
Target not recognised	Poor target, glare, distance or lighting.	Image detail, lighting and target quality.
Model not visible	Wrong hierarchy, scale or target assignment.	Child relationship and transform.
No planes	Provider disabled, unsupported device or unsuitable environment.	Package settings, device support and surface.
Tap does nothing	Missing raycast manager, no hit or input mismatch.	Input, plane detection, raycast and prefab.
Build fails	Missing modules or incompatible settings.	Build modules, package compatibility and settings.

Version note: Unity menu names, package versions and build requirements can change. Verify steps against the version installed in your lab before changing project settings.

8. Viva Questions and Short Answers

1. What is AR vs VR?

AR overlays digital content on the real environment; VR immerses the user in a virtual environment.

2. What is XR?

An umbrella term for AR, VR and MR.

3. What is Unity?

A real-time engine for building interactive 2D, 3D and XR applications.

4. What is a GameObject?

A container in a Unity scene to which Components are attached.

5. What is a Prefab?

A reusable template for configured GameObject instances.

6. What does Transform contain?

Position, rotation and scale.

7. Why use Time.deltaTime?

To make movement depend on elapsed time rather than frame count.

8. What is Vuforia used for?

AR experiences using tracked targets, including image targets.

9. What is an Image Target?

A predefined image the AR system recognises and tracks.

10. Why do target features matter?

Distinctive, distributed details improve visual matching reliability.

11. Why might a checkerboard perform poorly?

Repeated patterns can be ambiguous even if many corners are detected.

12. What is AR Foundation?

Unity's common AR API layer for supported provider plug-ins.

13. What are ARCore and ARKit?

Google's and Apple's AR platforms, respectively.

14. What does AR Plane Manager do?

Detects and manages trackable planes when supported and configured.

15. What does AR Raycast Manager do?

Tests screen positions against AR trackables such as planes.

16. What is an anchor?

A tracked pose reference used to keep content associated with a real-world location.

17. What is SLAM?

Simultaneous Localisation and Mapping.

18. What is occlusion?

Nearer real or virtual objects correctly hide content behind them.

19. Why test on a real device?

Tracking, camera behaviour, performance and input can differ from Editor simulation.

20. What is the key difference in this lab between Vuforia and AR Foundation?

Vuforia is used for image-target tracking; AR Foundation is used for markerless surface detection and placement.

9. Mini Practical Exercises

Level	Exercise	Success criteria
Beginner	Create and rotate a cube with C#.	Smooth rotation; adjustable speed.
Beginner+	Create a material and apply it to a cube.	Material appears correctly.
Intermediate	Create a Vuforia Image Target and attach a model.	Model appears and follows the target.
Intermediate+	Show information or play audio on target-found.	Events respond to target found/lost.
Advanced	Detect a plane and place a prefab with a tap.	Object appears at the raycast hit pose.
Advanced+	Compare targets under different lighting.	Record which image features help tracking.

10. One-Page Revision Summary

Topic	Remember
XR terms	AR overlays; VR immerses; MR blends spatially interacting real and virtual content; XR is the umbrella.
Unity basics	Scene → GameObject → Components → scripts / assets → build.
Vuforia flow	Image target → camera recognition → tracking → 3D content relative to target.
AR Foundation flow	AR Session + XR Origin → plane detection → tap → raycast hit → prefab placement.
ARCore / ARKit	Platform-specific AR providers used through supported Unity packages.
Good target	Sharp, distinctive, high-contrast and well-distributed detail; avoid blur, glare, blank areas and repetition.
Troubleshooting	Check camera, licence, target hierarchy, provider settings, raycast, input and prefab references.
Advanced quality	Stable tracking, clear interaction, device testing, frame timing and appropriate assets.

Source basis: supplied SRMIST Unit 3 lab PDF, “Build Your First AR Apps in Unity” (ARCore / ARKit, AR Foundation, Vuforia Engine, target quality and tap-to-place). General VR concepts and broader Unity explanations are supplemental study context.